FMAM 2026 rainfall outlook

ECMWF SEAS5 initialized 1 February 2026 · calibrated with CHIRPS v3.0 · FMAM rainfall contribution domain

Research reconstruction — not an official EMI or ICPAC forecast

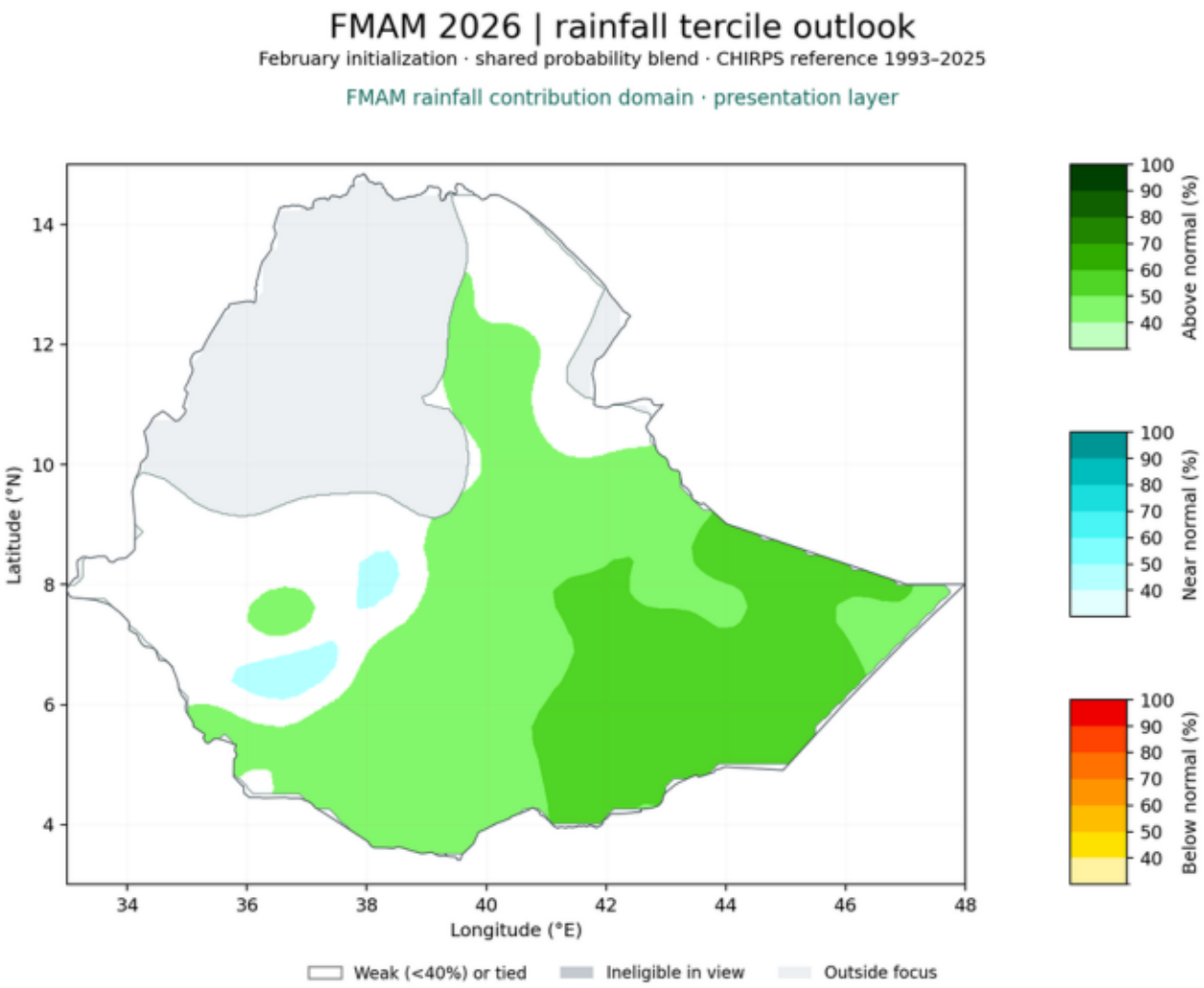
The forecast leans toward above normal rainfall in the FMAM rainfall contribution domain: averaged over the area, the local probability of above normal is 44%, against about 33% for climatology. Mean forecast rainfall is 347 mm against a 1993–2025 average of 310 mm (+37 mm, +12%).

Below normal 23% **Near normal 32%** **Above normal 44%**

Average local probability: each grid cell has its own tercile probabilities; these are their averages over the 100% of the area that has probabilities, not the probability that the area-total rainfall falls in a category.

Forecast signal: Moderate tilt toward above normal (largest area-average probability compared with a one-third reference; based on probabilities only).

Historical skill (FMAM rainfall contribution domain): Small estimated improvement; skill uncertain. Probability skill (RPSS) +0.017, cross-validated 1993–2016, 95% interval −0.038 to +0.067; better than climatology in 15 of 24 years. A strong signal is not a confidence rating.



Display only: 12× interpolation; Gaussian σ=1.5 native cells (≈0.38°, softens the 1° model grid). Statistics and exported fields use the original 0.25° grid.
Research reconstruction of a February-initialized forecast; not an official EMI/ICPAC product.

All targets, evidence and record

FMAM 2026 — February initialization · FMAM rainfall contribution domain

Target	Below/near/above	Coverage	Anomaly	Verification status
FMAM 2026	23% / 32% / 44%	100%	+37 mm	Verification published
Feb 2026	33% / 39% / 29%	91%	−3 mm	Verification published
Mar 2026	31% / 34% / 36%	100%	+7 mm	Verification published
Apr 2026	27% / 31% / 42%	100%	+32 mm	Verification published
May 2026	34% / 34% / 32%	100%	−2 mm	Verification published

Historical skill, cross-validated (probability skill, RPSS, with whole-year 95% interval)

Target	RPSS (95% interval)	Yrs better	Holm p	In words
FMAM	+0.017 (−0.038 to +0.067)	15/24	1.000	Small estimated improvement; skill uncertain
Feb	+0.137 (+0.050 to +0.234)	16/24	0.012	Moderate improvement over climatology (interval above zero)
Mar	+0.011 (−0.020 to +0.042)	15/24	1.000	Small estimated improvement; skill uncertain
Apr	+0.011 (−0.032 to +0.048)	16/24	1.000	Small estimated improvement; skill uncertain
May	+0.007 (−0.051 to +0.050)	17/24	1.000	Small estimated improvement; skill uncertain

Verification against CHIRPS (single season or month)

Target	RPSS	CRPSS	Bias	Forecast / observed anomaly	Observed below/near/above
FMAM 2026	+0.006	−0.081	+2.6 mm	+37 / +35 mm	13% / 40% / 47%
Feb 2026	+0.009	+0.028	−1.1 mm	−3 / −1 mm	33% / 45% / 22%
Mar 2026	+0.080	+0.081	−62.6 mm	+7 / +69 mm	0% / 4% / 96%
Apr 2026	−0.187	−0.648	+53.9 mm	+32 / −22 mm	45% / 39% / 16%
May 2026	+0.002	−0.031	+9.2 mm	−2 / −11 mm	37% / 46% / 17%

Cycle record

Model initialization: 1 February 2026 · ECMWF SEAS5 (system 51), 51 members

Target period: 1 Feb 2026 – 31 May 2026

Forecast record: Frozen 10 Oct 2026. Recalibrated with CHIRPS v3.0 when its observations for Feb 2026–May 2026 were already available (the fit itself uses 1993–2025 only): a retrospective reconstruction, not a forecast frozen before its outcome (SHA-256 manifest)

Publication status: Research reconstruction: produced after the initialization date with the published method; not an official EMI or ICPAC forecast

Reference period: CHIRPS v3.0 FMAM 1993–2025 (33 seasons)

Observations processed through: May 2026 (Feb, Mar, Apr, May)

CHIRPS archive: Official CHIRPS v3.0 monthly listing checked 12 Oct 2026: monthly files through Aug 2026

Verification last run: 11 Oct 2026

Method / version: Shared climatology blend (λ = 0.40 for FMAM) · code bed63b9

How to read: terciles split the reference-period rainfall into three equally likely parts. Climatology assigns about one-third to each (the observed tercile frequencies of the training years). Skill scores are decimals: +0.05 means a 5% lower score than climatology. RPSS refers to the blended probabilities; CRPSS and bias to the amount-corrected ensemble.

Limitations: few evaluation years and single verification seasons give wide uncertainty; the 0.25° maps repeat ~1° model information; the rainfall domain is for display and summaries only.

Monthly tercile outlooks

FMAM 2026 — February initialization · FMAM rainfall contribution domain

